When Does a Foundation Model Help? Task-Conditioned Benchmarking of Specialist, Zero-Shot, and Fine-Tuned MLIPs for Atomic Migration

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Abstract

Pre-trained foundation models may reduce the cost of building specialist machine-learned interatomic potentials (MLIPs), but their usefulness for a target process depends on distribution shift, not equilibrium accuracy alone. We introduce a candidate provenance-aware, distribution-aware model-selection framework demonstrated on one Cr-doped Sb_2Te_3 system, using from-scratch and foundation-based MACE models. We separate near-equilibrium pointwise error from process fidelity using fixed-path migration energies, self-relaxed NEB stability, collective interlayer sliding, and molecular-dynamics stability tests, with trajectory-grouped splits and seed-aware audits. The models have similar near-equilibrium errors but diverge on process tests. On the audited in-gap path, three-seed fine-tuning gives a fixed-path barrier error of $\mathbf{0.47 \pm 0.12}$ eV relative to the current provisional DFT reference, compared with $\mathbf{1.20 \pm 0.27}$ eV for grouped from-scratch training and $\mathbf{0.68 \pm 0.20}$ eV for a 5% scratch control; the historical **0.16** eV estimate is not reproduced. Zero-shot foundation checkpoints show target-specific barrier miscalibration, while three canonical from-scratch MD branches cross a predefined stability gate near 200 ps under the tested protocol. Higher-distortion paths remain an unresolved extrapolation test because converged DFT references and path-level seed statistics are incomplete. Our evidence supports a conditional decision framework: a foundation prior can help after task-relevant adaptation, but no model class is universally reliable. Model selection should follow task difficulty and distribution shift, with active learning providing a route to close the remaining evidence gap.

Keywords: Machine-learned interatomic potentials (MLIPs), foundation models, fine-tuning, defect migration pathways, 2D materials

1 Introduction

Machine-learned interatomic potentials (MLIPs) [1–14] provide a practical route around the accuracy–cost trade-off in atomistic simulation. Large pre-trained "foundation models," including CHGNet, [15] MEGNet, [16] M3GNet, [12] MACE-MP, [17] MatterSim, [18] and UMA, [19] have strengthened the expectation that broader data and scale can produce useful priors across many materials. [20] For an experimentalist, however, the decision is not whether a model is broadly capable in the abstract. It is whether the model reproduces a particular process in a particular material: here, the migration of a dopant through a layered semiconductor.

This creates a task-conditioned version of the specialist–generalist question. A specialist trained from scratch can be well calibrated to a target dataset, but its potential may become unstable when a molecular-dynamics trajectory reaches configurations that were not represented during training. A foundation model supplies a broader prior, but its energy reference, force scale, equilibrium volume, or potential-energy-surface curvature need not match the target DFT protocol. Target adaptation can reduce this calibration gap, but transfer learning can also introduce negative transfer when the downstream process lies far outside the representation supported by the foundation data [21]. A model that is useful for a within-gap migration event need not be reliable for a gap-to-QL/into-QL or within-QL event, or for collective sliding. Thus, the central problem is not to identify a universally superior model class, but to measure when a prior helps, when target data are required, and when the task remains unresolved.

Generic benchmarks such as Matbench, [22] JARVIS-Leaderboard, [23] and LeMatGenBench [24] establish broad utility for equilibrium properties and related benchmark distributions, but they do not by themselves resolve this process-level choice. Recent process-focused studies have shown the promise of foundation MLIPs for migration-barrier screening and diffusion, and of transfer learning for adapting foundation models with limited target data [21, 25, 26]. However, these studies do not establish a universal ordering of from-scratch, zero-shot, and target-adapted models for a common material and a common set of process observables. The relevant failure modes include target-specific miscalibration, potential-energy-surface softening, loss of long-range registry, and unstable force feedback during path optimization or MD. [27–34] Fine-tuning is therefore neither a universal cure nor merely a smaller-data version of pretraining; its value must be measured against task-relevant configurations and explicit stability gates. Uncertainty calibration may help identify high-error configurations for active learning, but it does not replace process-level validation [35]. Key questions of data efficiency, [36, 37] stability, [38, 39] and transferability [40, 41] therefore remain open for non-equilibrium processes. [42–44]

A rigorous benchmark must specify both the configuration distribution and the observables being measured. Conventional MLIP validation typically evaluates pointwise energy and force errors near equilibrium; migration and collective-displacement pathways test the same potential energy surface away from equilibrium. Figure 1 summarizes our audit sequence: near-equilibrium error, fixed-path calibration, self-relaxed NEB or MD stability, and hard out-of-distribution response. The ordering of tasks in

the schematic is a process-oriented design axis, not a claim that task difficulty is a single calibrated scalar. We therefore separate equilibrium pointwise accuracy, fixed-path energetic fidelity, and self-relaxed path stability in Sec. 2.3.

To demonstrate and validate this candidate framework, we employ doped two-dimensional (2D) van der Waals (vdW) materials as a controlled test bed. The discovery and design of novel 2D vdW materials [45–49] continues to drive innovation in spintronics, [50] quantum devices, [51–53] and energy technologies. [54, 55] A particularly powerful strategy for tuning the properties of these layered materials involves doping, the insertion of "foreign" atoms between vdW layers. [45, 56] This approach has enabled advances in materials design, ranging from lithium-ion batteries [57, 58] to the engineering of magnetic semiconductors [59] and quantum anomalous Hall systems. [60] Crucially, their layered structure offers distinct environments for atomic motion, making them a useful single-system test bed for comparing MLIP behavior across task-relevant configurational regimes.

Among 2D materials, topological insulators such as antimony telluride (Sb_2Te_3) [61, 62] represent an especially promising class for doping studies. The doping of transition metals has proven to be effective across multiple topological systems: Iron in Bi_2Se_3 creates ferromagnetic order [63–65]. Manganese in Bi_2Te_3 enables tunable magnetic properties. [66, 67] And vanadium in Sb_2Te_3 modifies its electronic structure. [68, 69] The doping of chromium (Cr) into Sb_2Te_3, the focus of this work, offers a compelling route to engineer its magnetic and topological properties for spintronic applications. [70–72] Predicting the stability and dynamics of these complex systems is essential for guiding experimental synthesis, a task for which large-scale Molecular Dynamics (MD) simulations are extremely powerful [73, 74] provided that a suitably representative force field is available. However, the atomic-scale processes governing functionality, including dopant migration and thermal stability, occur on time- and length- scales far beyond the reach of first-principles methods like density functional theory (DFT). [75–78]

This work demonstrates the candidate framework for Cr-doped Sb_2Te_3 within the current MACE model family (Fig. 1). The material supplies a compact task ladder: near-equilibrium configurations, a controlled within-gap migration path, gap-to-QL/into-QL and within-QL candidates, and collective interlayer sliding. We evaluate four training strategies—training from scratch, zero-shot foundation inference, single-temperature fine-tuning, and multi-temperature fine-tuning—using near-equilibrium errors, fixed-path migration energies, self-relaxed NEB behavior, MD stability under the tested protocol, and exploratory representation diagnostics. The benchmark separates two questions that are often conflated: whether a model is calibrated on a common reference path, and whether its own forces can stably generate a path or trajectory. It is therefore a case-study framework rather than a universal ranking of MLIP architectures or materials. As a supplementary scope diagnostic, three Cr-trained broad-roster checkpoints were evaluated on archived Cu/Li/Na paths in the same host. The MLIP outputs are finite and checkpoint-resolved, but all ten archived DFT path records retain SCF-convergence warnings; these results are therefore descriptive stress-test evidence rather than cross-element transfer validation. The present manuscript keeps generalization beyond the Cr-doped Sb_2Te_3 case as an explicit future validation target.

For numerical context, the LiF study reports activation energies of approximately 0.22 eV for MACE-MPA-0, 0.20 eV after fine-tuning on 300 configurations, and 0.24 eV for DeePMD; these values illustrate a transport-calibration question that is complementary to the present fixed-path barrier and self-relaxed-stability operators.

Three recent studies provide complementary benchmarks but address different questions. Alghamdi et al. [25] compare MACE-MPA-0 fine-tuning strategies for Li diffusion in LiF using nanosecond-scale molecular dynamics and Arrhenius transport observables, showing that a few hundred targeted configurations can reproduce a DeePMD reference. Bheemaguli et al. [26] evaluate zero-shot foundation MLIPs coupled to NEB over 574 battery-relevant migration paths, and separate barrier accuracy, conductor classification, and local transition-path geometry; this establishes broad screening utility without comparing specialist and fine-tuned training strategies in a chemically controlled system. Deng et al. [34] identify systematic underprediction of energies, forces, and PES curvature in universal MLIPs across out-of-distribution tasks, and show that sparse high-energy fine-tuning can correct part of this bias. We therefore treat PES softening as an established mechanism that motivates our diagnostics, not as a phenomenon first introduced here. Our contribution is to connect these strands in one doped layered material by comparing training strategies with fixed-path energetic fidelity, self-relaxed NEB stability, local migration, collective interlayer sliding, representation diagnostics, and provenance-aware evaluation.

This manuscript substantially extends our earlier non-archival AI4Mat workshop report [79]; it is not a duplicate publication. The earlier report introduced migration as a probe and provided an initial four-strategy MACE case study. The present journal manuscript adds a formal separation of fixed-path scoring from model-in-the-loop NEB optimization, distinct in-gap and higher-distortion migration probes, interlayer sliding, representation and surrogate-error diagnostics, grouped/provenance-aware splitting with nearest-neighbor overlap audits, and targeted multi-seed and data-volume controls. These additions support a narrower and more useful claim: migration pathways turn model selection into a measurable question about calibration, stability, and distribution shift, while the answer remains indexed to the pathway, training provenance, model initialization, and convergence status.

The completed evidence supports three observations. First, similar near-equilibrium errors do not imply similar process fidelity. On the audited in-gap path, the grouped three-seed fine-tuned model has a lower fixed-path barrier error than the grouped from-scratch model relative to the current provisional DFT reference, and the 5% from-scratch control is intermediate; the historical 0.16 eV point estimate is not reproduced after leakage-controlled retraining. Second, zero-shot foundation checkpoints provide a broad prior but show target-specific barrier miscalibration, so a smooth or self-consistent model path cannot be treated as a DFT-calibrated path without a common-reference check. Third, the canonical from-scratch MD branches cross a predefined scientific stability gate near 200 ps under the tested protocol, while the exact onset time varies across reruns; this is evidence of a stability risk, not evidence for a unique physical tipping point. The higher-distortion path family remains an extrapolation stress test rather than a completed model ranking because its converged DFT references and path-level seed statistics are not yet closed.

To this end, we address the following task-conditioned questions:

1. When equilibrium metrics are comparable, how do specialist, zero-shot foundation, and fine-tuned models differ in calibration, path stability, and long-trajectory stability?
2. Over what range of distribution shift does target-specific fine-tuning improve process fidelity, and where does a harder extrapolation task remain unresolved?
3. Can provenance-aware non-equilibrium diagnostics, including representation and force-disagreement analyses, identify where additional data or active-learning queries are needed without being mistaken for causal physical explanations?

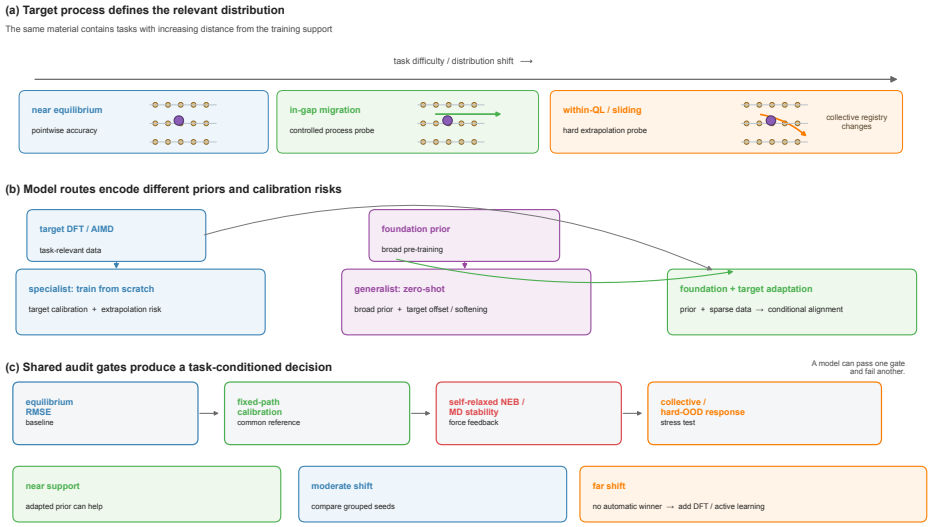


Fig. 1: **Task-conditioned model selection under distribution shift. (a)** A single Cr-doped Sb_2Te_3 system contains tasks with increasing distance from the training support, from near-equilibrium configurations through controlled within-gap migration to gap-to-QL/into-QL, within-QL, and collective-sliding probes. **(b)** Specialist, zero-shot foundation, and fine-tuned foundation routes expose different calibration and extrapolation risks. **(c)** A shared audit sequence separates equilibrium error, fixed-path calibration, self-relaxed path or MD stability, and hard out-of-distribution response. The bottom rule is a decision aid rather than a universal ranking: target-adapted priors can help near the measured distribution, while larger shifts require matched comparisons and additional DFT or active-learning data.

2 Methodology

We employed a systematic approach to benchmark various machine-learned interatomic potentials (MLIPs) for Cr-doped Sb_2Te_3[1], comparing specialist models trained from scratch against fine-tuned foundation models.

2.1 Data-Generation and Model-Training

Reference data were generated using density functional theory (DFT) calculations adopting the well-used PBE functional [76] and comprising roughly 20,000 configurations of a $4 \times 2 \times 1$ system of the Sb_2Te_3 crystal [80]. We use a 1 fs timestep for a series of *ab initio* molecular dynamics (AIMD) simulations at three temperatures (300 K, 600 K, 1200 K). We employed the MACE architecture for all models, which is a message-passing neural network for the representation of atomic interactions. [4]

This paper evaluates four distinct training strategies to benchmark their performance, named as follows:

1. **Scratch**: MACE model trained exclusively on our AIMD dataset
2. **Foundation**: Pre-trained MACE-MP model used without fine-tuning
3. **FT-600 K**: Target-adapted foundation model; the historical baseline used 5% of the 600 K trajectories, whereas the A2 leakage-controlled re-evaluation retrained the OMAT-FT-start family on the full trajectory-grouped training partition
4. **FT-MultiT**: Foundation model fine-tuned on multi-temperature data from the three temperatures studied

Revision 1 treats frame-level train/test splitting as insufficient for benchmark claims involving migration pathways. The revised data workflow assigns each entire source trajectory to a single partition (train, validation, or test) rather than shuffling individual frames, so that no trajectory is split across partitions. Under this grouped protocol (1,786/311/267 train/validation/test frames), the nearest-neighbor SOAP cosine distance between any audited fixed-path image and the grouped training set is 5.4×10^{-4}, and the closest comparable Cartesian RMSD is 2.06 Å; both are far outside the exact/near-duplicate thresholds used for this audit (SOAP $\leq 10^{-4}$, RMSD $\leq$ 0.05 Å), so no image/training-frame overlap was detected under the declared metrics. The Scratch-5% training subset (89 frames) gives the same result (SOAP 6.4×10^{-4}, RMSD 2.06 Å). Grouped-split retraining with three independent training initializations is complete for FT-600K fine-tuned from the OMAT-FT checkpoint, Scratch, and Scratch-5%, together with a single-checkpoint MACE-MPA-0 comparator (excluded from uncertainty estimates); the resulting RMSE and within-gap barrier statistics are reported below and in Table S1. Since a literal approximately 1000-frame control would fragment the source trajectories under this split, the current Scratch-5%

[1]This material system serves as an ideal testbed because its anisotropic van der Waals structure provides distinct migration environments: in-gap diffusion within quintuple layers (representing in-distribution scenarios) and interlayer migration across vdW gaps (probing out-of-distribution regimes). This structural duality enables efficient sampling of diverse configurations—from equilibrium states to high-energy transition states—while addressing technologically relevant processes where both thermodynamic stability and kinetic properties are critical for doping engineering.

result is reported as a grouped low-data control rather than as an exactly matched 1000-configuration ablation.

The reported fine-tuned models are documented as being trained on AIMD frames with real DFT energies and forces, not on NEB-derived exports lacking real force labels. The revised data manifest records source trajectory, split assignment, energy/force key mapping, and checksums for model-training files. The provenance audit covers 9,163 frames across the historical and grouped exports; the only nine exact zero-force frames are explicitly labelled one-atom Cr/Sb/Te isolated-atom E_0 references, with zero unclassified and zero NEB-derived zero-force frames. The complete manifest and audit code are in release commit `f5c6d19`.

2.2 Evaluation Framework

Models were assessed through a hierarchy of measurement operators that separates near-equilibrium pointwise accuracy from process fidelity on dopant-migration, molecular-dynamics, NEB, and collective-sliding configurations. Static energy and force errors, molecular dynamics observables, fixed-path NEB scoring, self-relaxed NEB stability, interlayer sliding, and representation diagnostics were evaluated as distinct probes of the same candidate MLIPs. Detailed computational parameters are provided in the Supplementary Information.

2.3 Formal Benchmark Definition

The central object of this benchmark is not a single scalar RMSE, but a family of process-conditioned measurement operators applied to the same candidate MLIP. Let a configuration be denoted by

$$x = (\mathbf{R}, \mathbf{Z}, \mathbf{h}) \in \mathcal{X},$$

where $\mathbf{R}$ contains atomic coordinates, $\mathbf{Z}$ atomic identities, and $\mathbf{h}$ the simulation cell. A reference electronic-structure model defines an energy $E_{\mathrm{D}}(x)$ and force vector $\mathbf{F}_{\mathrm{D}}(x) = -\nabla_{\mathbf{R}} E_{\mathrm{D}}(x)$. A candidate MLIP M defines $E_M(x)$ and $\mathbf{F}_M(x) = -\nabla_{\mathbf{R}} E_M(x)$.

Definition 1 (Equilibrium and process-conditioned evaluation distributions) The conventional equilibrium test distribution is

$$\mu_{\mathrm{eq}}(x) \propto \exp[-\beta E_{\mathrm{D}}(x)]$$

restricted in practice to relaxed structures, short AIMD trajectories, or random splits of near-equilibrium DFT configurations. Its standard pointwise error is

$$\mathcal{E}_{\mathrm{pt}}(M; \mu_{\mathrm{eq}}) = \left(\mathbb{E}_{x \sim \mu_{\mathrm{eq}}} \left[w_E |E_M(x) - E_{\mathrm{D}}(x)|^2 + w_F \|\mathbf{F}_M(x) - \mathbf{F}_{\mathrm{D}}(x)\|^2 \right] \right)^{1/2}.$$

For a physical process τ, such as dopant migration or interlayer sliding, the benchmark distribution is a process-conditioned distribution μ_τ supported on a structured set of configurations generated by a path operator. This distribution may place substantial probability mass on strained, high-energy, low-Boltzmann-weight configurations that are rare under μ_{eq} but control kinetics and mechanical stability.

Definition 1 makes explicit why equilibrium RMSE can be accurate and still incomplete: it measures the average pointwise error under μ_{eq}, whereas barrier prediction, diffusion, and sliding depend on E_M and ∇E_M over regions sampled by μ_τ. The two tests therefore answer different mathematical questions.

2.3.1 Pointwise Probes and Process Probes

A pointwise probe evaluates a model on isolated configurations,

$$\mathcal{P}_{\mathrm{pt}}(M;\mathcal{S}) = \{E_M(x), \mathbf{F}_M(x) : x \in \mathcal{S}\},$$

where $\mathcal{S}$ is a fixed set of DFT-labeled structures. This includes static energy/force RMSE and RDF-preserving near-equilibrium MD checks. A process probe evaluates a functional of a correlated configuration sequence,

$$\Gamma = (x_0, x_1, \ldots, x_N),$$

or of a trajectory distribution generated by M. Examples include a migration barrier, endpoint asymmetry, NEB convergence, diffusion coefficient, thermal conductivity, and sliding periodicity. The benchmark therefore measures both local correctness,

$$E_M(x_i) \approx E_{\mathrm{D}}(x_i), \qquad \mathbf{F}_M(x_i) \approx \mathbf{F}_{\mathrm{D}}(x_i),$$

and functional correctness,

$$\Phi_M(\Gamma) \approx \Phi_{\mathrm{D}}(\Gamma),$$

where Φ is a task-level observable rather than a pointwise label.

2.3.2 Migration-Path Selection

Let $\mathcal{A} = \{a_1, \ldots, a_K\}$ be metastable dopant sites obtained from relaxed DFT structures or thermally sampled AIMD motifs. A migration task is an ordered pair

$$\tau = (a_s, a_t, c),$$

where a_s and a_t are initial and final sites and c records the chemical and geometric context, such as in-gap motion, penetration into a quintuple layer, or interlayer sliding. The selected task set

$$\mathcal{T} = \mathcal{T}_{\mathrm{ID}} \cup \mathcal{T}_{\mathrm{OOD}} \cup \mathcal{T}_{\mathrm{coll}}$$

is designed to include: (i) interpolation-like paths whose local environments overlap with AIMD training data, (ii) extrapolation-like paths that pass through strained or high-energy transition states, and (iii) collective displacements where local environments change weakly but long-range registry or periodicity matters.

For each τ, a path is initialized by interpolation or chemically constrained image construction,

$$\Gamma_\tau^0 = (x_0, \ldots, x_N), \qquad x_0 = a_s, \quad x_N = a_t,$$

and then optimized by a reference or model-specific NEB operator. A path is retained as a benchmark probe only if its endpoints, chemical identity, cell convention, and image ordering are auditable from the raw inputs and outputs.

2.3.3 Fixed-Path and Self-Relaxed NEB Operators

The apparent discrepancy between a fixed-path barrier in Fig. 3 and a self-consistent MLIP-NEB barrier in Fig. S3 is resolved by defining two different operators.

Reference DFT path.

For a task τ, the DFT NEB operator returns a converged reference path

$$\Gamma^*_{\mathrm{D},\tau} = \mathrm{NEB}_{\mathrm{D}}(x_0, x_N) = (x^{\mathrm{D},*}_{0,\tau}, \ldots, x^{\mathrm{D},*}_{N,\tau}).$$

The DFT barrier is

$$\Delta E^{\ddagger}_{\mathrm{D}}(\tau) = \max_{0\le i\le N} E_{\mathrm{D}}(x^{\mathrm{D},*}_{i,\tau}) - E_{\mathrm{D}}(x^{\mathrm{D},*}_{0,\tau}).$$

Fixed-path scoring.

The fixed-path MLIP operator evaluates model energies on the same DFT-relaxed images:

$$\Delta E^{\ddagger}_{\mathrm{fixed}}(M,\tau) = \max_{0\le i\le N} E_M(x^{\mathrm{D},*}_{i,\tau}) - E_M(x^{\mathrm{D},*}_{0,\tau}).$$

It measures whether M assigns the correct potential-energy profile to an externally supplied DFT path. The model does not choose or relax the path.

Self-relaxed MLIP NEB.

The self-relaxed operator lets the MLIP define the forces that optimize the band:

$$\Gamma^*_{M,\tau} = \mathrm{NEB}_M(x_0, x_N) = (x^{M,*}_{0,\tau}, \ldots, x^{M,*}_{N,\tau}),$$

followed by

$$\Delta E^{\ddagger}_{\mathrm{self}}(M,\tau) = \max_{0\le i\le N} E_M(x^{M,*}_{i,\tau}) - E_M(x^{M,*}_{0,\tau}).$$

This operator measures optimization stability and qualitative path robustness on the MLIP's own PES. Because $\Gamma^*_{M,\tau}$ and $\Gamma^*_{\mathrm{D},\tau}$ need not be the same curve in configuration space, $\Delta E^{\ddagger}_{\mathrm{fixed}}$ and $\Delta E^{\ddagger}_{\mathrm{self}}$ are not expected to agree. A low self-relaxed MLIP barrier is physically validated only after the path is checked against DFT, for example by DFT single-point energies on $\Gamma^*_{M,\tau}$ or by restarting a DFT NEB from the MLIP path.

2.3.4 Error Decomposition

The fixed-path barrier error is

$$\epsilon_{\mathrm{fixed}}(M,\tau) = \Delta E^{\ddagger}_{\mathrm{fixed}}(M,\tau) - \Delta E^{\ddagger}_{\mathrm{D}}(\tau).$$

Define the local energy residual and the model/reference maximum-image indices as

$$\delta_M(x) = E_M(x) - E_{\mathrm{D}}(x),$$
$$i_M^* = \arg\max_i E_M(x_{i,\tau}^{\mathrm{D},*}), \qquad i_D^* = \arg\max_i E_{\mathrm{D}}(x_{i,\tau}^{\mathrm{D},*}).$$

The error can then be separated into endpoint and saddle-region terms:

$$\epsilon_{\mathrm{fixed}} = \underbrace{\delta_M(x_{i_D^*}^{\mathrm{D},*}) - \delta_M(x_0^{\mathrm{D},*})}_{\text{endpoint-to-saddle energy calibration}} + \underbrace{E_M(x_{i_M^*}^{\mathrm{D},*}) - E_M(x_{i_D^*}^{\mathrm{D},*})}_{\text{maximum-image localization on the fixed path}}.$$

Thus a model can fail by shifting the endpoint energy, miscalibrating the saddle energy, or moving the apparent maximum along the path. For a self-relaxed NEB, an additional geometric term appears because the model samples a different path:

$$\epsilon_{\mathrm{self}\to\mathrm{D}}(M,\tau) = \left[\max_{x_i \in \Gamma_{M,\tau}^*} E_{\mathrm{D}}(x_i) - E_{\mathrm{D}}(x_0)\right] - \Delta E_{\mathrm{D}}^{\ddagger}(\tau),$$

which measures whether the MLIP-discovered path is also plausible on the DFT PES. In practice this term requires DFT single-point calculations or DFT NEB refinement on the MLIP-relaxed path.

2.3.5 Robustness and Generalization Claims

The benchmark supports claims at three levels, with different evidence requirements.

1. *Pointwise accuracy*: $\mathcal{E}_{\mathrm{pt}}(M;\mu_{\mathrm{eq}})$ is small, as measured by energy/force RMSE, RDF agreement, and short-time stability.
2. *Process fidelity*: $|\Phi_M(\tau) - \Phi_{\mathrm{D}}(\tau)|$ is small for a specified task τ, as measured by barriers, transport observables, sliding profiles, or DFT-path scoring.
3. *Robust generalization*: Φ_M remains accurate over a documented task family $\mathcal{T}$, with multi-path, multi-seed, OOD, and convergence checks.

The present work claims Level 2 evidence for selected Cr-doped Sb_2Te_3 tasks and proposes Level 3 as a benchmark design principle requiring broader validation across materials, migration chemistries, architectures, and random seeds. This distinction prevents a successful result on one path from being over-interpreted as universal transferability.

2.3.6 Benchmark Algorithm

Algorithm 1 Migration-centered MLIP benchmark

Define candidate models $\mathcal{M} = \{M_1, \ldots, M_L\}$ and reference method D.
Construct or audit equilibrium data $\mathcal{S}_{\mathrm{eq}}$ from DFT relaxations and AIMD.
Train or adapt models using documented training splits, seeds, and code hashes.
Measure pointwise errors $\mathcal{E}_{\mathrm{pt}}(M; \mathcal{S}_{\mathrm{eq}})$ and near-equilibrium observables.
Select migration and collective-displacement tasks $\mathcal{T} = \mathcal{T}_{\mathrm{ID}} \cup \mathcal{T}_{\mathrm{OOD}} \cup \mathcal{T}_{\mathrm{coll}}$.
for each task $\tau \in \mathcal{T}$ **do**
    Generate endpoint-verified initial image sequence Γ^0_τ.
    Compute or restart $\Gamma^*_{\mathrm{D},\tau} = \mathrm{NEB}_{\mathrm{D}}(x_0, x_N)$ until convergence.
    Score each M on the fixed DFT path to obtain $\Delta E^{\ddagger}_{\mathrm{fixed}}(M, \tau)$.
    Optionally run $\Gamma^*_{M,\tau} = \mathrm{NEB}_M(x_0, x_N)$ to assess self-relaxed stability.
    If $\Gamma^*_{M,\tau}$ differs from $\Gamma^*_{\mathrm{D},\tau}$, validate the MLIP path with DFT single points or a DFT restart.
    Record barrier errors, endpoint errors, maximum-force histories, convergence flags, RMSD/SOAP overlap, and checksums.
end for
Report model rankings with uncertainty over seeds, paths, and convergence status.
Interpret latent-space or SHAP diagnostics as supporting evidence, not as a replacement for task-level validation.

2.3.7 Applicability Assumptions and Acceptance Criteria

The framework is applicable when the process of interest can be represented by auditable endpoint states and a physically meaningful path or trajectory functional. It is especially appropriate for rare events, migration, sliding, stacking rearrangements, and other processes where a small set of high-leverage configurations controls the downstream observable. It is less complete when the dominant failure mode involves electronic excited states, bond-breaking chemistry absent from the reference functional, long-range electrostatics beyond the model architecture, or finite-size effects not represented in the chosen cell.

A benchmark task is accepted into the final manuscript only if the following checks are satisfied:

1. DFT endpoints and NEB images are reconstructable from raw inputs and outputs.
2. The DFT reference barrier is convergence-verified, or explicitly labeled as a current candidate value.
3. Fixed-path and self-relaxed NEB quantities are reported as distinct operators.
4. Model comparison includes seed sensitivity where training stochasticity can affect rankings.
5. Training/test leakage is checked with grouped splits or path-aware proximity metrics.
6. Path overlap is quantified with geometric RMSD and/or descriptor-space similarity.

7. Any MLIP-discovered low-barrier path is tested on the DFT PES before being interpreted physically.
8. Scripts, model identifiers, input files, convergence flags, and output checksums are preserved in the release manifest.

2.3.8 Physics and Data-Evidence Chain

The physical basis of the benchmark is transition-state theory: kinetic rates depend exponentially on activation free energies, so small barrier errors can lead to large rate errors,

$$k \propto \exp[-\Delta G^{\ddagger}/(k_{\mathrm{B}}T)].$$

Consequently, a model that has low average force RMSE near equilibrium can still be unreliable if it misrepresents the PES along rare-event pathways. The evidence chain used here is:

$$\begin{aligned}&\text{DFT/AIMD data} \rightarrow \text{trained MLIPs} \rightarrow \text{equilibrium validation}\\ &\rightarrow \text{path/process probes} \rightarrow \text{DFT ground-truth comparison}\\ &\rightarrow \text{representation diagnostics} \rightarrow \text{data-acquisition guidance.}\end{aligned}$$

This chain closes the benchmark loop: physical theory identifies the observables that matter, DFT supplies the reference PES, MLIPs are evaluated through measurement operators matched to downstream processes, and diagnostic analyses indicate which missing configurations should be acquired next.

2.4 SHAP Analysis

We employed a model-agnostic SHAP workflow to identify SOAP descriptors associated with MACE energy-error behavior. Local atomic environments were encoded using SOAP descriptors (5.0 Å cutoff, $n_{\max} = 4$, $l_{\max} = 4$), capturing species-resolved radial and angular interactions in Cr-doped Sb_2Te_3. For each MACE variant, gradient-boosted regressors (200 trees, learning rate 0.05, max depth 8) were trained to predict the absolute total-energy error on the audited 150-frame trajectory. TreeSHAP values were then computed for the fitted surrogate models, and surrogate fidelity was assessed with 5-fold cross-validation rather than in-sample R^2. We compared the top 50 influential SOAP features across models to identify shared versus model-specific sensitivities in the surrogate error model.

3 Results and Discussions

3.1 Equilibrium Properties: Foundation for Comparison

As a baseline, we first evaluated all four candidate models on the near-equilibrium DFT data used in conventional MLIP validation. In the notation of Sec. 2.3, this corresponds to estimating a pointwise error under μ_{eq}, the distribution of relaxed and thermally sampled configurations near local minima. All models satisfy typical energy and force RMSE thresholds (Table S1), and the corresponding parity plots show visually similar agreement with DFT across energies and forces (Fig. S1).

Low equilibrium RMSE establishes the equilibrium baseline. Migration, sliding, and rare-event kinetics are measured separately with fixed-path and self-relaxed pathway probes whose observables depend on the potential energy surface along physically motivated displacements.

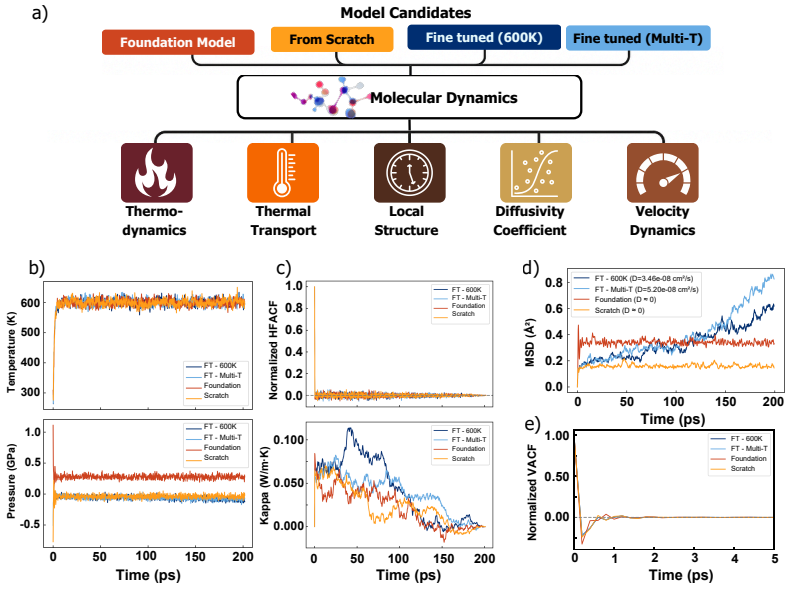


Fig. 2: **Comprehensive Benchmarking of MLIP Performance in Molecular Dynamics Simulations.** (a) A schematic of the unified evaluation protocol. Four candidate models (a zero-shot foundation model, a bespoke model trained from scratch, and two fine-tuned variants) were used to drive a 200 ps MD simulation. The resulting trajectories are then subjected to a uniform set of post-processing analyses to evaluate key physical properties. (b) Thermodynamic stability, demonstrated by the evolution of temperature and pressure over the simulation, remains near the target for all models. (c) Thermal transport properties, showing the Heat Flux Autocorrelation Function (HFACF) and its running integral to compute thermal conductivity (κ). (d) Mean-squared displacement (MSD), used to assess atomic mobility and calculate the diffusion coefficient. (e) Velocity autocorrelation function (VACF), which describes the underlying dynamics. Panels (b–e) use the same model mapping: dark blue, FT-600K; light blue, FT-MultiT; red, Foundation; orange, Scratch.

All MACE models (Foundation, Scratch, FT-600K, FT-Multi_T) successfully reproduce the equilibrium structure of Cr-doped Sb_2Te_3, with RDF analysis showing excellent agreement with the AIMD reference ("ground truth") across all atomic pair correlations (Fig. S2). This Figure includes a structural schematic of the local environment around Cr and full radial distribution functions (RDFs) for all models, confirming that each

training strategy preserves the characteristic nearest-neighbor and second-shell coordination features at 600 K. While all models maintain stable thermodynamic ensembles at 600 K (Fig. 2c), the zero-shot foundation model shows a persistent pressure offset, likely originating from its exposure to predominantly 0 K equilibrium structures rather than configurations extant at finite temperatures.

The models show similar RDF and VACF behavior but different transport estimates. The original FT-600K estimate used a shorter trajectory and a different sampling protocol; it is retained for provenance and excluded from model ranking. Under the current trajectories, the curves are best estimates rather than converged transport coefficients.

3.2 Non-Equilibrium Diffusion Pathways: The Critical Test of Generalization

The migration path benchmark evaluates a different measurement operator from the equilibrium RMSE. For each selected end-point pair $\tau = (a_s, a_t, c)$, the DFT NEB path defines a process-conditioned sequence of configurations $\Gamma^*_{\mathrm{D},\tau}$ that includes the metastable endpoints, intermediate distorted images, and the transition-state region. A candidate MLIP is then tested either by fixed-path scoring on this DFT geometry or by self-relaxed NEB optimization on its own PES, as defined in Sec. 2.3.3. These two operators answer different questions: whether the model assigns correct energies to the reference path, and whether the model can stably discover a plausible path without being given the DFT images.

3.2.1 Local Migration Events

For the fixed-path component of the benchmark, we used the DFT-relaxed NEB images as a common geometric reference and evaluated each MLIP by single-point energies on those images. This corresponds to $\Delta E^{\ddagger}_{\mathrm{fixed}}(M, \tau)$ in Sec. 2.3.3: the model is not allowed to move the path, so the resulting error isolates energetic fidelity along a DFT-supplied trajectory.

For the controlled within-gap probe, the provisional DFT barrier estimate is 0.34 eV. The archived QE NEB output has not yet passed the formal convergence gate; a changed converged value would replace the table entry and plotted DFT profile. The MACE Foundation fixed-path barrier is 1.02 eV, giving an audited fixed-path error of +0.69 eV relative to the provisional estimate. The Foundation profile is smooth but quantitatively miscalibrated on this reference path.

The Scratch profile overestimates the barrier, whereas the Foundation profile is smoother but remains quantitatively miscalibrated relative to the provisional DFT estimate. Fixed-path scoring and self-relaxed NEB are therefore reported as separate observables: the former tests energies on common DFT images, and the latter tests whether model forces produce a stable path.

We evaluated distinct Cr migration pathways (Fig. 3a) that sample different geometries. The primary reference probe remains within the vdW gap and is therefore an interpolative, process-conditioned case. The broader 61-atom path family is classified

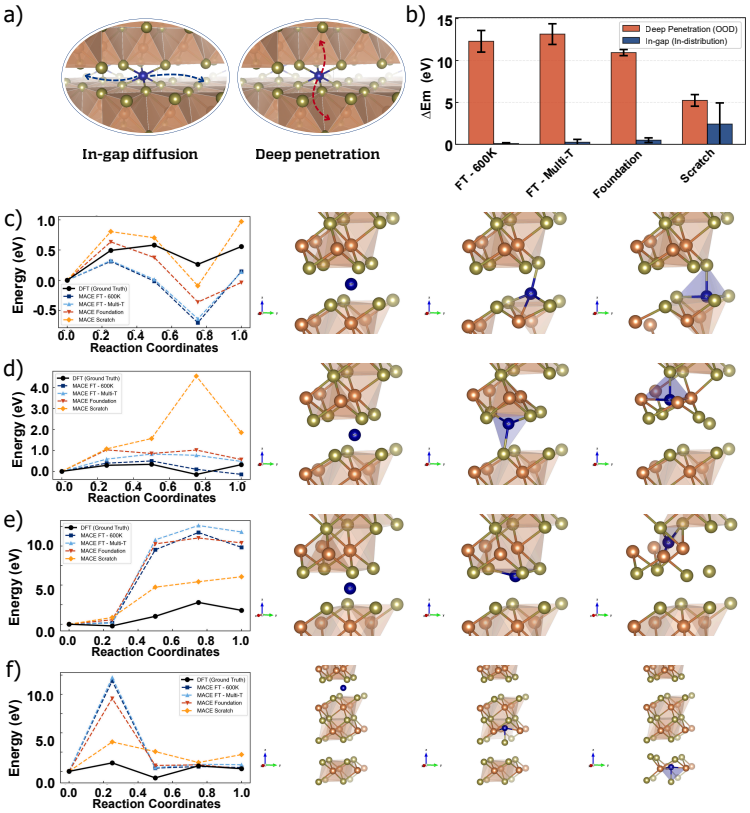


Fig. 3: **Comparative analysis of MLIP training strategies for predicting atomic migration barriers.** (a) Schematic illustration of the simulated Cr atom migration pathway between two stable sites within the Sb_2Te_3 bilayer. (b) Bar plot showing migration-energy prediction errors for the controlled within-gap probe and higher-distortion candidate classes. (c–f) Representative fixed-path energy profiles spanning the audited within-gap and gap-to-QL/into-QL regimes; internal audit identifiers are given in the Supplementary Information. In panels (c–f), black circles denote DFT, dark-blue squares FT-600K, light-blue triangles FT-MultiT, red down-triangles Foundation, and orange diamonds Scratch.

by physical regime in the Supplementary Information, including gap-to-QL/into-QL candidates, ambiguous transitions, and geometry-recheck cases.

In contrast, a within-QL event is defined by Cr entering the interior of a covalently bonded quintuple layer (QL), which distorts the surrounding lattice. Gap-to-QL/into-QL and within-QL paths form a separate higher-distortion OOD class.

In-Gap Diffusion: A Test of Interpolative Accuracy. For the within-gap diffusion comparison, the controlled probe is shown in Fig. 3d, while an additional within-gap candidate is retained as a separate diagnostic rather than pooled with it. Under the

grouped-split, three-initialization re-evaluation, the absolute fixed-path barrier error on the controlled probe relative to the provisional DFT estimate (0.34 eV) is 0.47 ± 0.12 eV for FT-600K fine-tuned from the OMAT-FT checkpoint, 1.20 ± 0.27 eV for Scratch, and 0.68 ± 0.20 eV for Scratch-5%; a single-checkpoint MACE-MPA-0 comparator gives 0.24 eV. FT-600K has the lowest reported error on this controlled probe. FT–MultiT was not retrained under the grouped protocol in this round.

The four representative foundation checkpoints (MP-0b3, MPA0, OMAT-0 medium, and MH-1) predicted barriers of 1.50, 1.79, 1.60, and 1.84 eV, respectively, on the same controlled within-gap images, compared with the provisional DFT value of 0.34 eV. All four overestimate the barrier by 1.17–1.50 eV; MPA0 is not the kinetic-accuracy winner in this completed comparison. Each checkpoint is reported individually because the foundation family does not provide training-seed replicates; full profiles and diagnostics are given in the Supplementary Information.

Higher-Distortion Candidates: A Test of Extrapolative Robustness. The gap-to-QL/into-QL and within-QL candidates provide a much more stringent test of model robustness. These cases involve significant lattice distortion as the Cr atom approaches or enters the covalently bonded quintuple layer (QL), creating atomic environments that are far from those seen in the training data (Fig. 3a).

The gap-to-QL/into-QL candidates sample larger lattice distortions and weaker overlap with the AIMD training distribution than the controlled within-gap probe. They form an OOD robustness probe. A final model ranking for this class requires the pathway classification and restarted QE references listed in the Supplementary Information audit; the within-QL regime remains unresolved.

The higher-distortion profiles vary across model classes and candidate regimes. In the grouped candidate audit, Scratch gives absolute errors of 0.51 ± 0.11 eV on one gap-to-QL/into-QL candidate, 0.72 ± 0.55 eV on a second, and 1.77 ± 1.00 eV on a geometry-recheck candidate; the first result is comparatively stable within the tested initializations, whereas the second shows greater run-to-run spread and the geometry-recheck candidate remains both high-error and seed-sensitive. These values replace a qualitative "by chance" explanation with evidence, but they do not establish a final hard-OOD ranking because the candidate DFT references and pathway classes are not uniformly closed. The broader matrix retains force-pathology flags and separate model lineages, so its rows are used as descriptive failure-mode evidence rather than pooled with the strict grouped comparison.

The Critical Role of Task-Specific Fine-Tuning

Fine-tuning improves process fidelity when adaptation data overlap with the pathway distribution. The FT-600K model, fine-tuned on data generated at the target temperature of the migration event, has the lowest reported error on the selected in-gap probe (Fig. 3b).

The 600 K fine-tuned model is closer to the controlled within-gap migration probe than the broader multi-temperature model in this case study. The grouped-split SOAP/RMSD audit finds no fixed-path-image overlap with either the full training set or the Scratch-5% subset. The Scratch-5% control gives a within-gap error of 0.68 ± 0.20 eV, between FT-600K (OMAT-FT start; 0.47 ± 0.12 eV) and Scratch (1.20 ± 0.27 eV).

This ordering is consistent with a useful foundation initialization under moderate distribution shift, but the 89-frame grouped control does not isolate a pure pretraining effect at exactly matched approximately 1000-frame data volume. FT–MultiT was not retrained under grouped protocol in this round, and the higher-distortion outputs are retained as class-resolved descriptive diagnostics in the Supplementary Information rather than pooled into a final ranking.

The 600 K and Multi-T results show that training-data selection changes process fidelity. Benchmark-guided acquisition can therefore target configurations that improve robustness on the measured pathways.

3.2.2 NEB Stability as an Additional Robustness Criterion

We distinguish the fixed-path operator from self-relaxed NEB because they measure different aspects of process fidelity. The historical 0.41 eV value previously shown in Fig. S3 is not retained as a quantitative self-relaxed barrier: the audit identifies it as a historical 82-atom, two-Cr system with twelve images, rather than the 61-atom, one-Cr system used in Fig. 3d, and finds inconsistent endpoint/internal calculators and corrupted final-band energy serialization. It is therefore retained only as a provenance failure audit and excluded from model ranking. A valid same-system self-relaxed comparison would require rerunning the MLIP NEB with matched endpoint/image identity and, ideally, DFT rescoring of the MLIP-relaxed images.

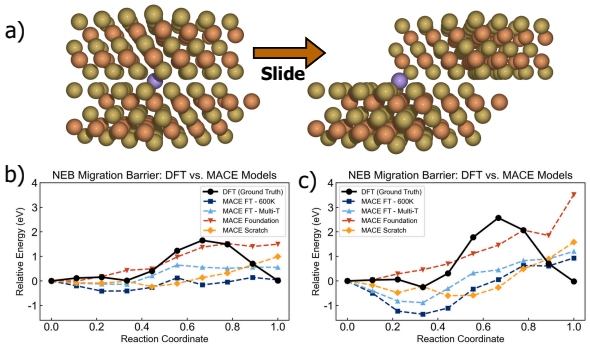


Fig. 4: **Evaluation of MACE models on collective lattice displacement: interlayer sliding in Sb_2Te_3.** (a) Schematic illustration of the bilayer sliding process in Sb_2Te_3, where the top layer (yellow/orange atoms) slides relative to the bottom layer along the crystallographic direction. The purple sphere indicates the position of a Cr dopant when present. (b) Energy barriers for interlayer sliding in pristine Sb_2Te_3 as calculated by DFT and the MACE models. (c) The corresponding barriers for Cr-doped Sb_2Te_3. Black circles denote DFT, dark-blue squares FT-600K, light-blue triangles FT-MultiT, red down-triangles Foundation, and orange diamonds Scratch in both panels.

3.3 Interlayer Sliding: Probing Robustness to Non-Local, Collective Displacements

Having tested the models' response to local, high-distortion events, we evaluated their robustness to a different class of out-of-distribution challenge: large-scale, collective lattice displacements. To this end, we simulated the shearing of one Sb_2Te_3 layer relative to another, a process governed by the subtle corrugations of the interlayer vdW potential energy surface (Fig. 4a). This task is particularly challenging for MLIPs as the local atomic environment of any given atom changes only minimally, while the global configuration undergoes a significant transformation that crosses periodic boundaries. The configurations along this sliding path were not present in the training data.

We first examined the case of pure, undoped Sb_2Te_3 (Fig. 4b). The Foundation model gives the closest barrier shape and magnitude among the models. Its final state remains higher in energy than the identical initial state, and Scratch shows the same endpoint drift.

The fine-tuned models (FT-600K and Multi-T) underestimate the barrier. Adaptation that improves local Cr chemistry does not preserve this collective sliding observable.

The same pattern appears in Cr-doped Sb_2Te_3 (Fig. 4c): all models miss barrier height and endpoint periodicity. The collective sliding probe therefore separates local bonding accuracy from long-range registry accuracy.

3.4 Representation Learning Analysis

To diagnose possible origins of the observed performance differences, we conducted a multi-faceted analysis of the learned representations. We projected high-dimensional atomic environment descriptors from each model's 600 K MD trajectory onto a two-dimensional space using two complementary visualization techniques: t-distributed stochastic neighbor-embedding (t-SNE) [81] to assess representational dissimilarity, and Potential of Heat-diffusion for Affinity-based Trajectory Embedding (PHATE) [82] to visualize trajectory geometry (Fig. 5).

The apparent separation in the t-SNE projection (Fig. 5a) is consistent with different recorded descriptor distributions for Foundation, Scratch, and the fine-tuned models, but it is not a direct mechanistic explanation. In the audited analysis, the original 6,191-dimensional Euclidean silhouette score is 0.33, the cosine score is 0.28, and the scale-controlled z-scored Euclidean score is -0.02; the projected baselines are 0.40 ± 0.03 for t-SNE across perplexities 5, 30, and 50 and 0.47 ± 0.00 for PHATE. The t-SNE silhouette spans 0.36–0.42 with trustworthiness 0.963–0.973; at fixed perplexity, the nominal seeds are identical under the locked PCA initialization, so this spread primarily reports perplexity sensitivity. The raw descriptor vector mixes energy, force, RDF, and SOAP components, and the reduction after scaling control shows that the original-space metric is descriptive and scale-sensitive rather than evidence for disjoint local-geometry manifolds.

The PHATE projection (Fig. 5b) provides a complementary visualization of the recorded time evolution, while its low-dimensional geometry is compared with potential energy in Fig. 5c. Along PHATE 1, specialist models occupy similar regions and

PHATE 2 separates Scratch from the fine-tuned models; these patterns are used as qualitative diagnostics and do not establish a physical mechanism.

These representation differences accompany the observed diffusion and NEB differences, but the causal relationship is not resolved by this analysis. The all-zero force-sensitivity stub was removed before the audited calculation, and no finite-difference force-sensitivity claim is made.

Latent-space separation accompanies the task-level differences, and we use it as a diagnostic alongside the barrier, transport, and stability benchmarks.

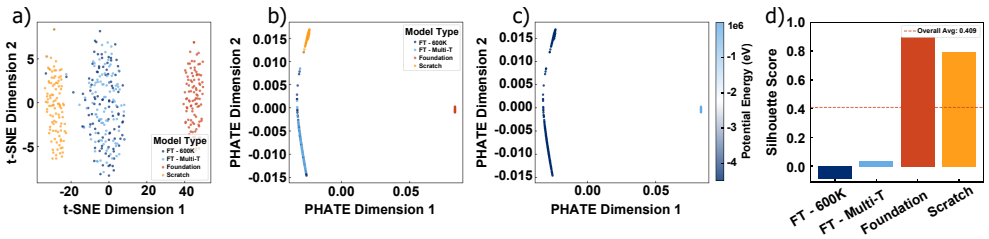


Fig. 5: **Representation diagnostics from trajectory embeddings and original-descriptor checks.** Projections of atomic environment descriptors from 600 K MD trajectories. **(a)** The apparent separation of Foundation (red), Scratch (orange), and fine-tuned models (blue) in the t-SNE projection is consistent with different recorded descriptor distributions, but is not a direct mechanistic explanation. **(b)** PHATE visualizes the recorded trajectory geometry; separation along PHATE 2 is a qualitative diagnostic whose sensitivity is reported in the SI. **(c)** The same PHATE embedding colored by potential energy shows a Foundation energy offset that is consistent with a scale or calibration mismatch, not by itself with a physical mechanism. **(d)** Silhouette scores compare original-space and projected representations; the original-space values (raw Euclidean 0.33 and z-scored Euclidean -0.02) are the primary scale-aware check, whereas the projected values are descriptive baselines.

3.5 Dissecting Learned Representations with Feature-Level Interpretability

While t-SNE and PHATE suggest that distinct training strategies produce separable descriptor patterns, these projections do not establish the physical principles learned by each model. To add a feature-level diagnostic, we use SHapley Additive exPlanations (SHAP) [83] on surrogate models of MACE's absolute energy error. This analysis identifies SOAP descriptors to which the surrogate error predictors are most sensitive; it does not directly explain the internal message-passing mechanism of MACE.

Our methodology is designed to identify descriptors correlated with predictive error. For each MACE model, we trained a Gradient-Boosting Regressor to predict absolute total-energy error from 390 SOAP and 17 structural descriptors extracted from the X-FORCE 150-frame trajectory; frame identity is joined explicitly through

`image_index`. [84] This approach interprets a simpler surrogate of the error landscape, not the internal message-passing mechanism of MACE itself. The audited 5-fold cross-validation R^2 values are 0.98, 0.88, and 0.97 for FT-600K, FT–Multi–T, and Scratch, respectively. We then applied TreeSHAP to these surrogate models, assigning feature-importance values to SOAP descriptors that are most correlated with prediction errors.

Our analysis shows a divergence in the surrogate feature-importance landscapes associated with each training strategy. A global overview of the top 50 influential descriptors across models (Fig. 6b) shows that the fine-tuned multi-temperature (FT-MultiT) surrogate concentrates importance more strongly, whereas the scratch-model surrogate distributes importance more diffusely. In the locked revised ranking, the largest FT-MultiT contribution is `Cr-Sb_n43_l0`; the leading Scratch contributions are dominated by Cr–Te descriptors, while Cr–Cr descriptors appear in some rankings without being a stable dominant channel. These are descriptive surrogate sensitivities, not evidence that the descriptors are causal or mechanistically internalized by MACE.

Figure 6a quantifies this disparity. The FT-MultiT profile is concentrated on Cr–Sb local-coordination descriptors, whereas the Scratch profile is broader. These feature patterns describe surrogate error predictors; they do not identify the physical mechanism of the NEB behavior. A complementary 846-task native-MACE endpoint perturbation campaign is reported in the SI as a completed local-stability diagnostic, but it is not an isolated Cr–Cr SOAP-channel perturbation and does not convert the SHAP association into a causal claim.

4 Conclusions

We benchmarked specialist and foundation-based MLIPs for Cr-doped Sb_2Te_3 using the measurement-operator framework defined in Sec. 2.3. This formalism separates equilibrium point-wise accuracy from fidelity on dopant-migration and collective-sliding probes, and it distinguishes fixed-path energetic scoring from self-relaxed path optimization. Under this framework, diffusion trajectories and structured pathways show that models with similar equilibrium metrics can diverge sharply under kinetically relevant distortions.

Our findings support several broader hypotheses. Robust benchmarking should span both in-distribution and out-of-distribution tasks, because accuracy near equilibrium does not guarantee kinetic fidelity. Fine-tuning can improve a selected process while reducing fidelity for other response regimes; this trade-off must be measured rather than assumed away. Latent-space analysis provides a qualitative diagnostic for why models with similar pointwise metrics may behave differently in downstream tasks. The higher-distortion path candidates remain outside the final numerical ranking until their geometry labels and restarted QE references are closed. The cross-element broad-roster audit provides a reproducibility and evidence-gating diagnostic, but its ten Cu/Li/Na DFT path records retain SCF-convergence warnings and therefore do not constitute transfer validation. Accordingly, the generality of the task-conditioned framework beyond this Cr-doped Sb_2Te_3 case remains an open validation question.

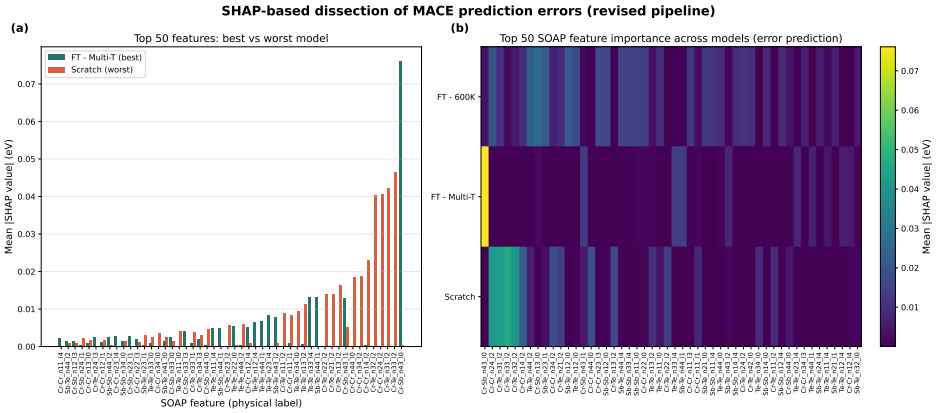


Fig. 6: **SHAP-based Dissection of Feature Importance in MACE Models.** **(a)** Mean absolute SHAP values for the top 50 SOAP features, comparing the best-performing fine-tuned model (FT-MultiT) and the worst-performing scratch-trained model. The corrected audit identifies `Cr-Sb_n43_l0` as the largest mean absolute SHAP feature in the FT-MultiT surrogate; the values describe sensitivity of a surrogate error predictor and are not causal attributions to MACE. **(b)** Heat map of surrogate feature importance for the top 50 SOAP features across all training strategies. The FT-MultiT surrogate concentrates importance on fewer features, while the scratch-model surrogate distributes importance more diffusely.

Together, these results indicate that adding "more data" alone is insufficient; pathway-relevant, non-equilibrium sampling can be decisive for meaningful improvement. Migration-inspired probes, coupled with latent-space diagnostics and auditable data provenance, provide a practical route to identify which configurations are needed and where models are likely to fail. In this case study, model selection is task-specific: the appropriate strategy must be validated against the particular non-equilibrium observables it is expected to predict. Claims of universally complementary model classes require validation across additional materials and architectures.

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- **Conflict of interest/Competing interests**
  The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.
- **Ethics approval and consent to participate**
  Not applicable. This study does not involve human participants or animal subjects.
- **Consent for publication**
  Not applicable.
- **Data availability**
  The supporting dataset, including training manifests, NEB path metadata, model-predicted barriers, and post-processing tables, will be released through MigrationBench on Hugging Face at https://huggingface.co/datasets/alinacao2000/MigrationBench. The manifest records the source trajectory, split assignment, random seed, model provenance, DFT/MLIP task type, convergence status, and checksum for each derived artifact. NEB paths are evaluation/reference data and were not merged into the fine-tuning datasets. The frame-level force audit covers 9,163 released training frames and found no unclassified or NEB-derived all-zero-force frame; the nine exact zero-force records are explicitly labelled isolated-atom E_0 references (release commit `f5c6d19`).
- **Materials availability**
  Not applicable. This work is entirely computational and does not involve the synthesis or distribution of physical materials.
- **Code availability**
  The scripts used for data processing, DFT/MD post-processing, and plotting are available on GitHub https://github.com/yicao-elina/MigrationBench.git. The Sec. 3.5/Fig. 6 SHAP chain is released as `scripts/shap_analysis.py`, `scripts/shap_feature_labels.py`, and `scripts/plot_shap_fig6.py`, using the compact three-model X-FORCE inputs in `data/raw_compact/revision1/shap/`. The source scope and MD5s are recorded in `docs/audit/shap_source_lineage.md`; the four-model, 2-concentration, and Dual-X ID/OOD/NEB workflows are not inputs to Fig. 6.
- **Author contribution**
  Y. C.: conceptualization, methodology, data curation, DFT and MD simulations, analysis, visualization, and manuscript-writing. **P. M.**: literature review, conceptualization, and manuscript- writing. **P. C.**: conceptualization, project administration, supervision, funding acquisition, and manuscript-writing.

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